

Cambridge International AS Level
Information Technology 9626 · Paper 1 Theory

Topic 9

Modelling

17 questions from 16 question papers, May/June 2022 to May/June 2026.

Questions come first, in date order, then their mark schemes in the same order. Each question is labelled with the paper it came from. Click "Mark scheme" next to a question to jump to its answers, or use the bookmarks panel.

A question that covers more than one topic appears in each of those topics. Topics follow the 2025-2027 syllabus.

Questions

9626/11 May/June 2022 · Question 2

[Mark scheme](#)

2 Flight simulators are used to train pilots to fly modern jet aircraft.

Describe a typical flight simulator.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

..... [6

..... [5]

Analyse the effectiveness of using a spreadsheet model for this purpose.

[8]

(b) The organisers wish to achieve a larger profit.

By explaining what each step requires them to do, describe how the organisers would use goal seek to find the size of the audience needed to make a profit of \$5000. You should refer to specific cells from the spreadsheet in your answer.

Include in your answer a reason why the result might not be satisfactory.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

9 Computer models are often used to run simulations when designing a nuclear power plant.

(a) Describe the benefits of using simulations in this scenario.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

(b) Describe the drawbacks of using simulations in this scenario.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

[8]

[6]

[8]

[8]

[6]

[8]

[8]

[8]

Discuss the benefits and drawbacks of using computer simulations in this context.

[8]

4 Identify and explain **two** drawbacks of using a spreadsheet for computer modelling.

1

.....

.....

.....

.....

2

.....

.....

.....

.....

[4]

8 Identify and describe **three** characteristics of modelling software.

characteristic 1

.....

.....

.....

.....

characteristic 2

.....

.....

.....

.....

characteristic 3

.....

.....

.....

.....

[6]

4 Computer models make use of what-if analysis.

(a) Describe, with the use of relevant examples, how what-if analysis can be used as part of a model of population growth.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

- (b)** Evaluate the use of a computer simulation as a suitable method to plan a response to a natural disaster.

[8]

[8]

..... [2]

Mark schemes

Mark scheme · 9626/11 May/June 2022 · Question 2

[Back to question](#)

Question	Answer	Marks
2	Four from: Simulates difficult situations/take off scenarios/speed of the aircraft/turbulence/different/extreme weather conditions/can control weather conditions Include software and specialised equipment/hardware... ...that can cost hundreds of thousands of pounds... ...a cabin containing screens, flight controls, control wheel, throttle, joystick, instrumentation, pilot seat, sound system, computer hardware, altitude gauges, fuel gauges (must have two) Software includes the rules of the model that define how the aircraft reacts... ...includes a database containing information about each airport... ...allowing pilots to practice landing at unfamiliar airports... Control panel in a cockpit mimics/is identical to one in a real plane/equipment is designed to react in a similar way to real aircraft Pilot uses joysticks/levers/switches/buttons/sliders to control movement/speed Displays are generated by computers on the screens in the cockpit/screens display runway/sky... ...computer graphics produce display using... ...projectors/back-projection screen/panoramic spherical mirror	4

Mark scheme · 9626/11 May/June 2022 · Question 7

[Back to question](#)

Question	Answer	Marks
7	Six from: Replication/copying means you don't have to enter the same formula/data several times/the rules of a model can be changed easily using replication Absolute and/or relative cell referencing makes sure you only increment the parts of a formula you need to when replicating a formula Allows what-if questions to be asked Cell protection makes sure that the cells you want do not change by accident User interface forms makes it easier to input values into the model Conditional formatting allows you to highlight certain values that match specific criteria Graphs/charts to show trends/predictions The use of input forms makes it easy to enter data Macros make it easier to create more complex formulae or functions/used to record a sequence of actions Goalseek can be used to determine which variables need to be changed to achieve a target or goal Pre-written functions allow users to create complex formulae Values can be changed to ask whatif questions	6

Mark scheme · 9626/13 May/June 2022 · Question 7

[Back to question](#)

Question	Answer	Marks
7	Five from: Testing nuclear reactor designs using computer models avoids safety problems... ...such as explosions/meltdowns <u>Testing nuclear reactor designs</u> using computer models cuts costs... ...as don't have to pay as much money for workers/materials to replace damaged reactors Can obtain results in a short period of time regarding reactions that take a long time in real life Re-designing computer models is cheaper than re-building a nuclear reactor Researchers will have the tools to simulate scenarios that are hard to observe in operating reactors It may be too dangerous to modify the actual nuclear reactor You can test predictions more easily/model can make predictions more accurately	5

Mark scheme · 9626/11 Oct/Nov 2022 · Question 7

[Back to question](#)

Question	Answer	Marks
7	Eight from: • Spreadsheet models can do complex calculations (1) • Formulas are recalculated automatically ... (1) ... speeding up the use of the model (1) • Graphs that are produced to show population growth ... (1) ... will (automatically) change as new values are added/old values altered (1) • (Reliable because) provide consistent results from the same inputs //will always produce the same answer (1) • Spreadsheets can automatically download data to provide data in the fields (1) • Cannot account for every possible variable when forecasting population growth ... (1) ... therefore model can give misleading results (1) • To try to account for as many variables as possible requires complex formulae ... (1) ... which may need experts who are expensive to hire (1) • Producing an effective model may be time consuming (1) • The model can have formulae replicated (1) • Complex population predictions may need additional software/modules ... (1) ... which is expensive to buy (1) • Data can be entered accurately because of computer-based validation and verification (1) Max. six marks if bullets/list of points Max six if no expansions	8

Question	Answer	Marks
4(a)	Three from: • They could increase/change (the number of customers by increasing/changing) the value in <u>cell D1</u> ... (1) • They could increase/change (the number of snacks by increasing/changing) the value in <u>cell C6</u> ... (1) (accept decrease answer) ... • They could increase/change (the number of drinks by increasing) the value in <u>cell C5</u> ... (1) (accept decrease answer) ... • They could increase/change (the cost of souvenir supplies by increasing) the value in <u>cell D17</u> ... (1) (accept decrease answer) The value in D21/profit would change (showing the new profit/loss//profit would change) (1)	3
4(b)	Four marks available Max three from: • The organisers would select the cell containing the formula that returns the result they want/D21, and select goal seek (1) They input the value they want the formula to return/5000 (1) They enter the cell reference of the input value that the spreadsheet can change to reach the target/D1 (1) Max one from: • The number could be to several decimal places and needs to be a whole number of concert goers (1) • If other assumptions are incorrect (e.g. the number of snack/drinks bought per concert goer), the number produced by goal seek would be inaccurate (1) • Venue may not be able to hold that many people (not 'number too large' on its own) (1)	4

Question	Answer	Marks
9(a)	Four from: • Definition of a simulation e.g. Simulations are a digital representation of a situation as it would happen in real life (1) • Using computer models avoids safety problems (1st) • such as e.g. explosions/meltdowns (1) • Using computer models cuts costs (1st) • as don't have to pay as much money for workers//materials to replace damaged reactors (1) • Can obtain results in a short period of time regarding reactions that take a long time in real life (1) • Re-designing computer models is cheaper than re-building a nuclear power plant (1) • Can simulate scenarios that are hard to observe in operating power plants//in real life (1)	4
9(b)	Four from: • Simulation may not be 100% accurate (1) • Models can't always recreate exactly the real-world experience/may miss variables (1) • Very complex to develop a simulation (1) • Specialist knowledge is required to build model (1st) • which requires specialists which costs money (1) • Specialist knowledge is required to analyse results (1st) • which requires specialists which costs money (1) • There may be too many variables to include in the model (1st) • to exactly simulate the environment/design/working of the nuclear reactor (1) • leading to inaccurate results (1)	4

Question	Answer	Marks
11	Eight from: • Weather forecasting used to involve several weather forecasters and days of calculations (1st) • today one model can manage the task (1) • Computer models help forecasters to make more reliable predictions (1) • Models can be used to predict future patterns (of weather) (1) • More complex models create more accurate forecasts (1st) • which allow forecasts further into the future (1) • Human weather forecasters are still needed to evaluate the results (1) • Weather forecasting requires the use of a lot of data (1) • To be effective models must consider many variables (1st) • such as (TWO Variables identified): atmospheric pressure, humidity, rainfall, temperature, wind speed, wind direction (1) • as well as calculations such as (ONE of): vertical air motions, temperature changes, moisture processes (1) • A lot of computing power is needed to run a computer model so supercomputers have to be/are used (1) • Supercomputers are (very) expensive (so makes the process an expensive one) (1) • Models can be used to compare current conditions with previous similar conditions/shown trends over a period of time (1) Max 6 marks for just bullet points	8

Question	Answer	Marks
1	Six available. - Spreadsheet software allows what-if analysis to be carried out (award any mention of what if analysis) One per instance of a suitable characteristic: e.g. - Can sort data into a rank order - Can protect cells from accidental change - User interface forms to allow data input Can use macros which can create routines /subroutines - Macros/subroutines enable automation//can be executed by a button - Conditional formatting can show/highlight data that matches//is greater than//less than criteria - Lookup tables to store values - Automatic recalculation changes output when new value inputted/data updated - Allows values to input (in bulk) from an external source - Carries out calculations such as subtraction (accept any example of a calculation) - Formulae can be created for calculations - Functions can perform set tasks in formulae - Goal seek finds a suitable starting input for a desired output - Charts to present values//allow for comparison - Charts automatically update	6

Question	Answer	Marks
6	Eight from: Advantages: MAX SIX • The running/maintenance costs of a flight simulator are less than an actual aircraft (1) • Avoids repair costs for real aircraft (1) • Different/various/a range of situations can be simulated in a flight simulator (1st) – planned at the start of the simulation (1) – during the simulation (1) • Events can be simulated without waiting for them to happen in real life (1st) – such as (suitable named example) (1) • Situations can be repeated multiple times until the pilot gets it right (1) • (Safer because) Situations can be practiced in a flight simulator without putting the humans/trainee/flight training instructor in danger (1) • Flight simulator does not need to be at an airfield//can be in a convenient location (1) • Can concentrate training on specific controls (1) • The environment benefits from flight training in a simulator (1st) – as there is no pollution created by a flight simulator (1) • The level of noise in a flight simulator is not as loud as in an aircraft (1st) – makes the communication between the instructor and trainee much easier (1) Disadvantages: MAX FIVE • Simulators are not as realistic/real as a real aircraft (1) • Pilots get overconfident and may compromise safety (1) • It is difficult to recreate boredom/fatigue of a real flight (1) • Available simulated scenarios dependant on software packages (1) • Pilot may “freeze” when faced with a real flight/actual passengers (1)	8

Question	Answer	Marks
6	SIX marks available e.g. • Computer is used to create the rules behind a simulation (1st) so that the simulation matches the real world (1) • Easier to try out a range of different scenarios//allows what-if analysis (1) • It is less costly than if a mistake is made in real life/it is costly to remedy mistakes (1) • More difficult to construct several designs in real life (1) • Can be used to check on/make predictions for complicated scenarios//scenarios with a lot of variables (1) • Can observe aspects that would be difficult to observe in real life (1) • Can save time (1st) more tests can be done in a shorter amount of time (1) a much greater number of tests can be performed (1) • Simulation modelling can be used to plan for infrequent/one off events (1) • Computer models are much safer than experimenting in real life (1) • Faster, can speed up variables that would impact on results e.g. climate change (1)	6

Question	Answer	Marks
11	EIGHT marks available. Benefits (max 7): e.g. • Can provide results that are generally difficult to measure (1st) – such as e.g. reaction times. (1) • Can simulate a range of scenarios (1) • Some scenarios do not happen frequently in real life (1) • Can simulate driving in different vehicles. (1) • Reduces anxiety level//improves confidence of (new) learners. (1) • Can be repeated any number of times. (1) • Can be customised/personalised to the learner's needs. (1) • Simulation can be recorded (1) • Can analyse mistakes (1) and suggest solutions (1) • Simulation can be frozen to allow for questions/discussion (1) Drawbacks (max 7) e.g. • May not have enough data/information to create some scenarios. (1) • May not accurately predict the occurrence/effects of human behaviour. (1) • Might provide misleading scenarios if information was incorrectly/erroneously inputted/added. (1) • Can require high-end hardware for optimal performance increasing price. (1) • May give drivers a false sense of security regarding their ability to drive. (1) • Not a real-world experience (1)	8

Question	Answer	Marks
11	Eight marks from: MAX four for 1st marks only <i>Evaluations: E.g.:</i> • Can interrogate/interact with databases to extract data (1st) – So data only has to be typed/entered in once (1) • Can only use the features of the spreadsheet (1st) – Therefore model may be limited (1) • Spreadsheet software may include unneeded features (1st) – Which could be a distraction//which need to be hidden//increase file size (1) • Spreadsheets include formulae//functions (1st) – So complicated calculations can be created more easily (1) – Reduced chance of error (1) • Calculations can be performed more quickly/more easily/recalculated automatically (1st) – so spreadsheet can be used by someone who does not spreadsheet editing skills (1) • What-if statements can be asked/edited (1st) – without rebuilding the model/spreadsheet from scratch each time (1) • Creates graphs (1st) – Which allows data to be more easily interpreted (1) • Graphs that are produced will automatically change (1st) – as new values are added/old values altered (1) • Models/spreadsheets provide objective results (1st) – not affected by user's bias (1) • The use of templates provides consistency over time (1st) – therefore do not need to relearn the model (1) • Valid conclusion drawing together/based on arguments presented in the answer (1)	8

Question	Answer	Marks
13	Max one for: - Definition of computer simulation (1) Max six from: <i>Benefits</i> - Identified cost saving (1) - Drivers are not put in physical danger/can be safer/less risk of being hurt/no risk to life (1) - Different/extreme conditions can be simulated so no need to wait for natural occurrence (1) - Conditions/scenarios can be repeated//each simulation can be reset to same starting point/values/circumstances (1) - Simulation can be recorded for later playback/analysis (1) - Feedback can be visual overlays of previous test (1) - Real time data/telemetry produced (1) - Computer can more accurately measure performance metrics (1) - Greater number of testers could be used (1) - Testing can occur at multiple points in the development process (1) - Can try different (possible) features of the car without having to actually build it (1) Max six from: <i>Drawbacks</i> - Can be difficult to simulate all conditions found on roads (1) - Can be expensive to create a realistic simulation (1) - Lack of a realistic experience of the car (1) - Drivers may be more reckless in simulations because of lack of real danger (1) - Could produce phantom/ghost/erroneous problems (1) - Significant analysis/expertise may be required to make sense of data (1) - Simulated scenarios may be inaccurate/wrong (1)	8

Question	Answer	Marks
4	Four available. Max two for each matched pair: e.g. - Impossible/difficult/improbable to include all the possible variables (1st) - too many/unknown/unexpected variables (variables in widest sense) (1) - Spreadsheet software is limited size/capacity/scalability (1st) - plus suitable explanation of an issue this causes (1) - May not have the required formulae/functions (NOT “options”) (1st) - plus suitable explanation of an issue this causes (1) - Need training to use the software (properly)/requires an expert/complicated for new users (1st) - plus suitable explanation of an issue this causes e.g. cost of expertise (1) - therefore takes time to create (do not award as extension for “requires an expert”) (1) - Spreadsheets can be edited by anyone who has the software (1st) - So different versions may develop (1) - So different outcomes arrived at (1)	4

Question	Answer	Marks
8	SIX marks available: ONE mark per identified characteristic, ONE mark for the description/purpose of that characteristic. • Can carry out WHAT-IF analysis (1) So that scenarios can be explored (may exemplify scenario being changed)(1) Allows for complexity (1) • Has inputs/outputs (1st) So that values can be set by the user//so that results can be seen by the user (1) • Can create rules (1st) These define how the inputted data/values are used//the calculations that take place (1) • Can edit rules (1st) To take account of changing parameters/conditions (1) • Cell protection (1st) Protects cells from being (accidentally) changed by the user (1) • User forms (1st) Allows the easy input of data (1) Allows data to be entered in an organised way (1) • Can create macros (1st) To automate processes (1) • Automatic recalculation (1st) Outputs automatically updated (1) • Can hold/handle large amounts of data (use of Multiple – award as BOD).(1st) To allow for different possible outcomes/scenarios (1) Allows for complexity (1) (may be repeated from MP 1) Note: – NOT DEPENDANT • Includes function/formulae (1) to perform calculations/carry out equations (1)	6

Question	Answer	Marks	Guidance
4(a)	FOUR marks available, two marks per matched and described example. • What–if analysis explores what will happen if something else occurs/didn't occur/a variable is changed (1) • Rate of birth can be changed (1) to see impact of increased babies (1) • Rate of mortality can be changed (1) to see impact of increased//decreased teenagers//young adults/adults on number of births (1) • Life expectancy can be changed (1) to see impact of people living longer//shorter on overall population. (1) • Change structural constructs (e.g. schools, hospitals, access to water) (1) to see impact on e.g. mortality rates (1)	4	

Question	Answer	Marks	Guidance
4(b)	EIGHT marks available: Simulations can model/represent what happens in the real world/a natural disaster (1) Positives (max 6) • Simulation can be used in advance of a disaster (1) • Do not have to wait for the disaster to happen//simulations can run now (1) • Simulations can show (possible) impacts/damage of the disaster (1) • Identifies aspects that can be planned e.g. evacuation routes (1) • (e.g.) buildings/infrastructure can be checked to check resilience (1) • Structural/environmental changes can be made (based on the results) (1) • The replacement cost of a e.g. collapsed bridge exceeds the cost of a simulation (1st) Therefore money is saved by using on a simulator (1) • Simulations can repeat scenarios//allow for trial and error//can test different plans out (1st) e.g. So the response team can explore different responses//check if their response works (1) Learn from their mistakes (1) • There is no risk to human life in a simulation (concept of reduced risk/not being in a risk situation) (1st) Therefore the simulation is safer (1) • Small changes can be made to the scenario//different scenarios can be checked/explored/used (1st) To research into marginal effects (concept of) (1) To allow the team to practice in changing conditions/explore the impact of small changes (1) • Simulations can be recorded (1st) for (e.g.) future analysis/training (1)	8	

Question	Answer	Marks	Guidance
4(b)	Negatives MAX 6 • Simulations may not include all possible events (1) Simulation is not realistic (1) • Simulations assume fixed responses/fixed rules (1st) People may respond differently in real life (1) • Initial cost of purchase of software may be expensive (1) • Interpretation of the outcomes requires experts//team members from different groups (1) • Interpretation of the outcomes requires time//has a cost implication (1) Conclusion (may only be awarded if positives(s) and/or negatives are included) • One mark for a value judgement/conclusion based on two arguments (1)		

Question	Answer	Marks	Guidance
6(a)	Eight from: • Means does not use a (real) helicopter (1st) avoids wear and tear//maintenance costs on the helicopter (1) • Simulations copy the real world (1st) • So, pilots can experience relatively rare/dangerous (1) A range of real-life situations (1) Which may not otherwise occur//may have to wait for (1) <u>For dangerous ONLY and if safety not already awarded</u> – “avoids the risk” (1) • Concept of injury being avoided (1st) Therefore, the simulation is safer (1) • The running cost of a simulation is less than the running cost of a helicopter (1st) And so, money is saved//is a cost-effective solution/is cheaper (1) • It replicates the experience of being in a real/different helicopter (1st) by having the same layout/controls (as a real helicopter) (1) • The replacement cost of a (crashed) helicopter exceeds the cost of a simulation (1st) Therefore, money is saved by training on a simulator (1) • Simulations can repeat scenarios (1st) So, the trainee pilot can master key skills//learn from their mistakes (1) • Small changes can be made to the scenario (1st) To allow the pilot to practice in changing conditions (1) • Different locations can be trained for (1st) Without the need to go there (1) • Pilots can train on their own in the simulator (1st) So, no need for a trainer to be present (1) More lessons can take place (at the same time) (1) • Lesson can be saved (1st) And content analysed afterwards (1) MAX six marks if the candidate only gives first mark answers.	8	

Question	Answer	Marks	Guidance
6(b)	TWO marks available for an explanation of an identified disadvantage: • Pilot may have a false sense of security (1st) And so take more risks in a real helicopter (1) • Simulator does not risk anyone's life, but a real flight does (1st) So pilot may freeze when faced with really being in charge (1) • Concept that may not be totally accurate or (fully) represent real life or realistic (1st) Therefore not a value practice (1) May not inspire//give confidence in the real world (1) Cause problems when they experience things that were not practiced (1)	2	ONLY mark first drawback given/attempted Do Not Award • Training is inefficient – TV • Isn't up in the air • The pilot could get scared on a real flight • Is a less pressured experience (TV) • A simulation is not training on real life (as in, it is a simulation, not the real thing). This is different to “a simulation may not be realistic”. • It doesn't feel like flying a helicopter